MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BVOC (HONOURS) Regular EXAMINATION

THIRD SEMESTER

MG3MPCREM200 - Introduction to solar cells and applications (2025 ADMISSION ONWARDS)

Duration: 1.5hr **Maximum Marks:50**

Rememder(K) , *Understand(U), Apply(A), Analyse(An),* Evaluate(E), Create(C), Skill(S), *Interest(I) and Appreciation(Ap)*

PART A

Short answer

Answer any 15 questions

Each question carries 1 marks

1. Define the term solar constant. [CO1- K, U]
2. Identify the main components of solar radiation reaching the Earth's surface. [CO1- K, U]
3. State the approximate wavelength ranges of solar radiation. [CO1- K, U]
4. Mention any two limitations of solar energy. [CO1- K, U]
5. Identify the semiconductor material commonly used in commercial solar cells. [CO2-U, An]
6. Define the photovoltaic effect in a solar cell. [CO2-U, An]
7. State the function of the PN junction in a solar cell. [CO2-U, An]
8. Define the fill factor of a solar cell. [CO2-U, An]
9. Identify the semiconductor material used in CdTe thin-film solar cells. [CO3-U, An]
10. State the full form of CIGS used in thin-film solar cells. [CO3-U, An]
11. Mention the absorber material used in perovskite solar cells. [CO3-U, An]
12. Identify one emerging solar cell technology based on organic materials. [CO3-U, An]
13. Identify the type of solar PV system commonly used for electricity generation in solar farms. [CO4-A, Ap, S]
14. State one application of solar energy in residential buildings. [CO4-A, Ap, S]
15. Mention any two off-grid solar applications. [CO4-A, Ap, S]

16. Define a grid-connected solar PV system. [CO4-A, Ap, S]

17. Identify one major component of a grid-connected solar PV system. [CO4-A, Ap, S]

PART B

Short Essay

Answer any 3 questions

Each question carries 5 marks

18. Explain the properties of solar radiation and distinguish between direct and diffuse radiation. [CO1- K, U]

19. Discuss the solar spectrum and explain the different regions of the solar spectrum. [CO1- K, U]

20. Explain the working principle of a solar cell based on the photovoltaic effect and PN junction formation. [CO2-U, An]

21. Compare crystalline silicon and thin-film solar cells with respect to materials, construction, efficiency, advantages, and applications. [CO3-U, An]

22. Explain the applications of solar energy in homes and buildings, solar-powered lighting and gadgets, and off-grid systems such as solar pumps and chargers. [CO4-A, Ap, S]

PART C

Essay

Answer any two questions

Each question carries 10 marks

23. Explain solar energy in detail, including its advantages, limitations, major applications, and its role in meeting future energy demands. [CO1- K, U]

24. Explain the I–V characteristics of a solar cell and discuss the efficiency and various factors affecting its performance. [CO2-U, An]

25. Explain the different types of solar cell technologies, including crystalline silicon, thin-film (CdTe, CIGS, and a-Si), and emerging technologies such as perovskite, organic, and dye-sensitized solar cells, highlighting their working principles, advantages, limitations, and applications. [CO3-U, An]

26. Explain the working and applications of grid-connected and off-grid solar PV systems, including solar panels for homes and buildings, solar farms, solar-powered lights and gadgets, and solar-powered pumps and chargers. [CO4-A, Ap, S]